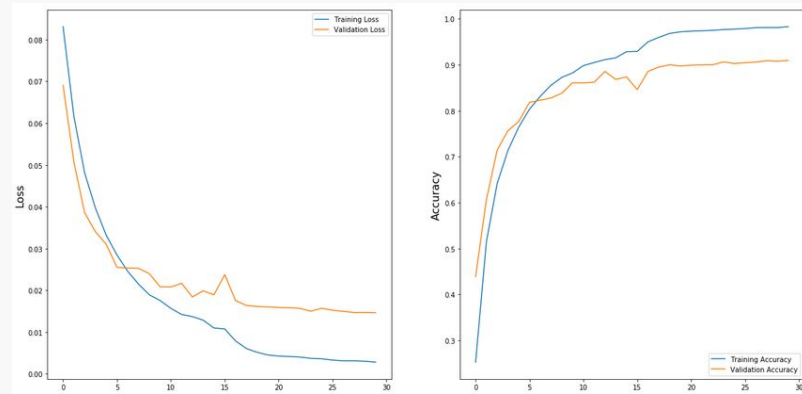


Optimized Vision Transformer (ViT) Training Pipeline for CIFAR-10

Ayush Tripathi, Sai Minnal, Mayank Deoras

Problem Statement & Motivation

- Deep learning projects often optimize for final accuracy
- Final accuracy by itself doesn't fully represent efficiency
- Compute cost & hardware efficiency matter
 - Particularly with diminishing returns
- We are working to minimize time required for a ViT to reach 85% test accuracy on CIFAR-10



*Standard accuracy and loss curves for model training against epochs
— clear diminishing returns*

Technical Approach: Objective + Evaluation Protocol

- Task: CIFAR-10 image classification

$$(x \in \mathbb{R}^{32 \times 32 \times 3}, y \in \{1, 2, \dots, 10\})$$

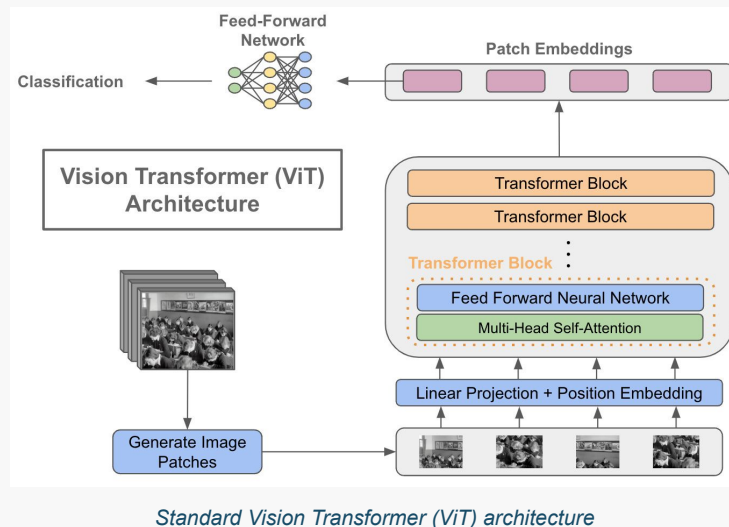
- Goal metric: time-to-target accuracy (target $\tau = 94\%$)

$$T_{\tau}(\theta) = \min\{t \geq 0 : A_{\text{test}}(t; \theta) \geq \tau\}$$

- Optimization Problem:

$$\theta^* = \arg \min_{\theta} T_{\tau}(\theta)$$

- What is θ — model hyperparameters and training method
- Metrics logged — wall-clock time, test accuracy each epoch, throughput (images/sec), peak GPU memory
 - Early stop once $A_{\text{test}} \geq \tau$



Mathematical Formulation of Approach

- ViT breaks images into N patches (P by P) $\rightarrow$ feature vector (vector of length d)

$$N = \left(\frac{H}{P}\right) \left(\frac{W}{P}\right)$$

- ViT adds positional embeddings (attention doesn't know order/location)

$$Z_0 = [z_{\text{cls}}; z_1; \dots; z_N] + E_{\text{pos}}$$

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

- QK^T is an N by N matrix — attention cost grows like $O(N^2)$
 - Patch count N is a huge performance indicator

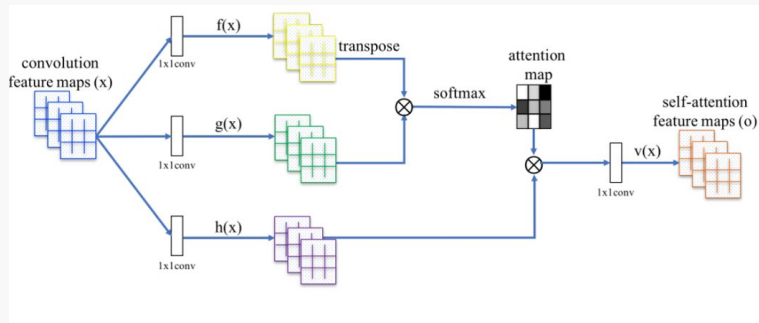


Illustration of the self-attention mechanism

Methodologies

Non-deterministic

- Training run isn't exactly reproducible each time
- Fixed seeds aren't used → enhanced training time

Automatic Mixed Precision (AMP)

- Uses lower precision arithmetic where safe to speed up computation and decrease memory usage

Random Augmentation

- Applied randomized image transformations during training
- Adds extra computer per step → increases data diversity

Learning Rate Schedulers

- Adjusts LR during training to improve time to convergence
-

Additional Optimizations

torch.compile

- Graph-level kernel fusion, reduces Python overhead

Muon Optimizer

- Orthogonalized gradient updates for 2D transformer weights
- AdamW handles biases, embeddings, LayerNorm

Mixup + Label Smoothing

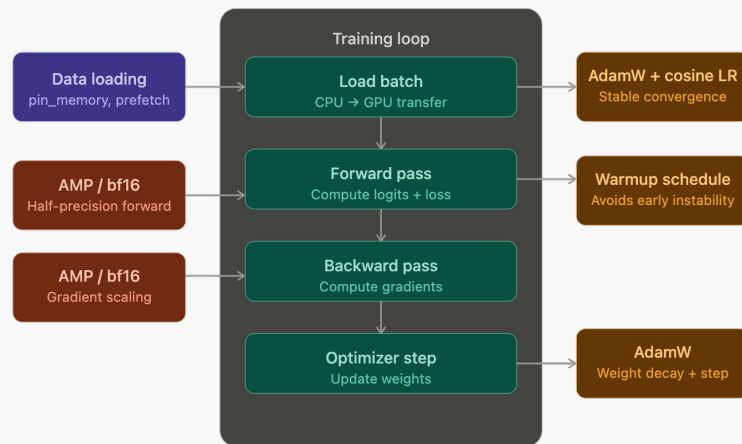
- Combines images/labels per batch, softens hard targets ($\epsilon = 0.1$)

BF16 Precision

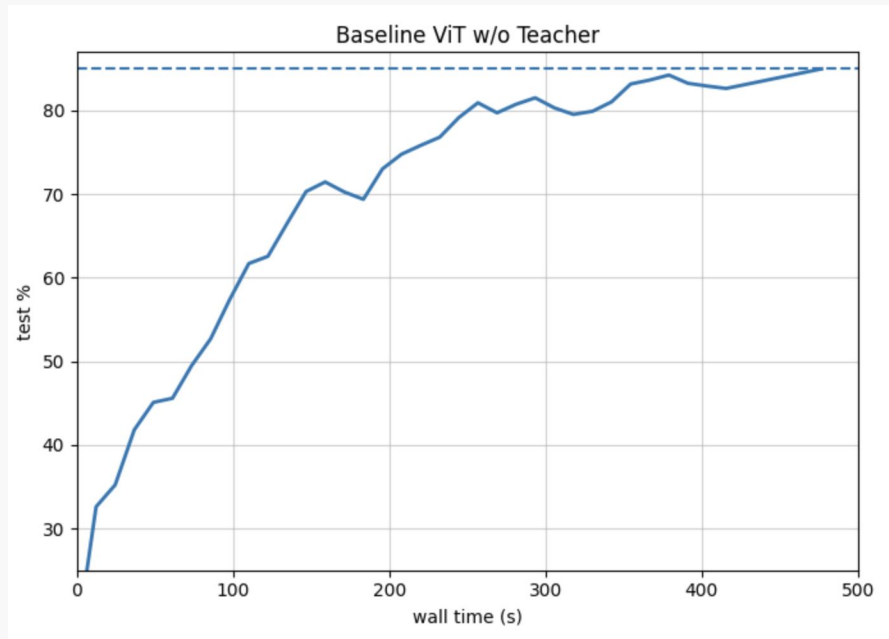
- More stable than FP16, lower memory on A100

Global Average Pooling

- Replaces CLS token, averages patch representations
- Better convergence stability on small datasets



Initial Results After Optimizations



Baseline: reached 85.0% in 476.7s

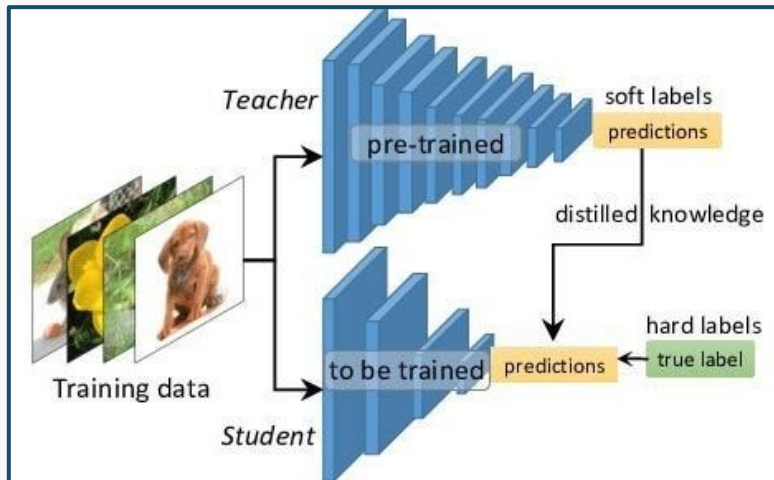
Pivoting to Distillation

Insight

System-level optimizations improved speed per step.

Convergence behavior is not optimized.

Using a ViT allows for more flexibility and power at scale compared to a CNN, but takes longer to train.



New Goal:

Minimize time required to reach 85% test accuracy on CIFAR-10 using various distillation methods.

Pivot

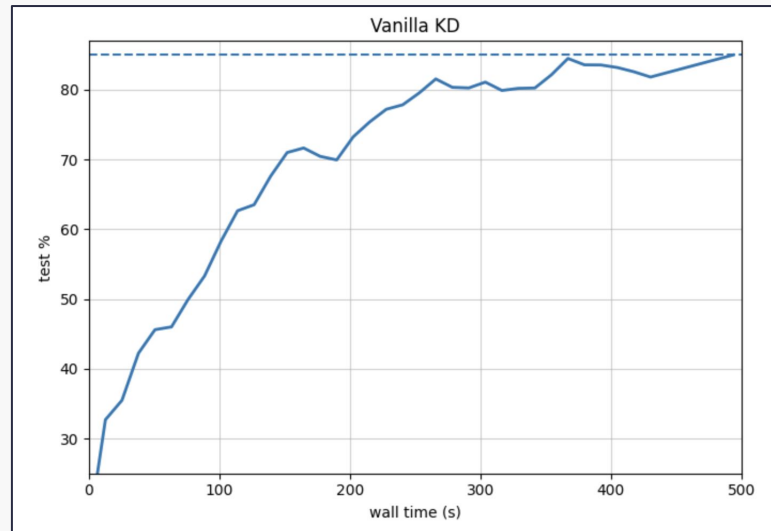
Bottleneck isn't just the system but how the model learns.

One-Hot Encoding leads to limitations.

CNN teacher-guided training provides richer, more informative supervision with faster convergence while retaining the benefits of the ViT.

Vanilla Knowledge Distillation (KD)

- **Setup**
 - Teacher (ResNet-56, frozen) → Student (ViT)
 - Train on labels and teacher predictions
- **Key Idea**
 - Combine standard cross-entropy loss with hard labels and distillation loss between student and teacher logits
 - Teacher provides soft targets (probability distributions)
 - Captures class relationships (“dark knowledge”)
- **Benefits**
 - Richer supervision than labels alone
 - Faster convergence and smoother training
- **Limitation**
 - Treats all non-target classes the same



KD reached 85.0% test accuracy in 493.6s

Decoupled Knowledge Distillation (DKD)

- **Setup**

- Same teacher–student setup as KD
- Distillation is applied on teacher and student outputs

- **Key Idea**

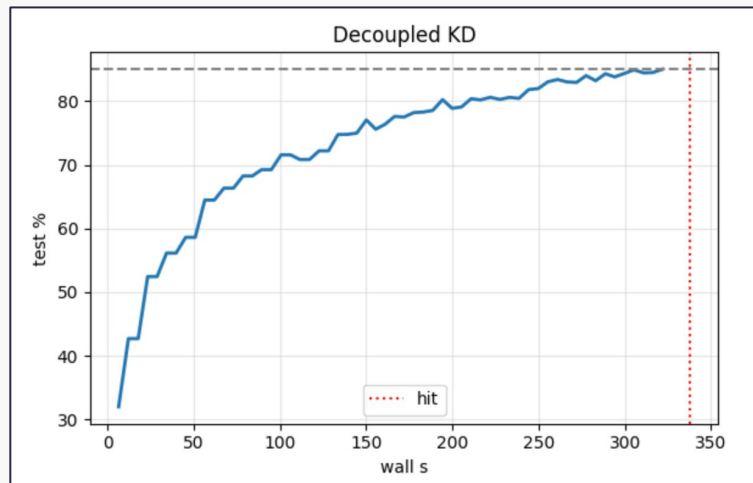
- Distillation is split:
 - TCKD (target class alignment)
 - NCKD (non-target “dark knowledge”)
- Controlled by hyperparameters:
 - α (target emphasis), β (non-target emphasis), and temperature (T)

- **Benefits**

- Separates correctness vs class relationships
 - Better control over learning dynamics
 - More stable and effective training

- **Limitation**

- Adds extra hyperparameters
- Requires tuning for best performance



DKD reached 85.0% test accuracy in 337.81s

Next Steps

- Implement feature-level distillation
 - Align intermediate representations not just logits
- Select appropriate transformer layers for feature alignment
- Measure convergence gains from feature matching
- Evaluate computational overhead of feature matching
- Refine DKD hyperparameters further